FreAuth: Novel Frequency Feature-Based Device Authentication for Magnetic Wireless Charging

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Abstract—**Device authentication plays a crucial role in preventing illegal access and ensuring smooth usage of magnetic wireless charging. However, current authentication techniques suffer from security vulnerabilities and are incompatible with low-cost receiver devices, thus severely limiting their applications. In this paper, we propose `FreAuth`, a novel Frequency feature-based device Authentication technology for magnetic wireless power transfer systems. Technically, we begin by conducting circuit measurements at the transmitter side to subtly retrieve impedance information related to the receiver without the need for its corporation. Then, we employ a dual-frequency interleaved-based subtraction technique to remove the ideal receiver impedance and capture the fairly weak frequency features. Furthermore, we normalize the captured frequency features to account for environment variations. These steps allow us to generate and store a hardware fingerprint for the receiver based on its frequency features. During device authentication, we use a discrete Fréchet distance-based algorithm for fingerprint matching. We devise and implement a prototype of `FreAuth` and conduct extensive experiments to evaluate the proposed scheme. The experimental results validate the reliability (95.74% authentication accuracy among 60+ devices) and robustness (anti-interference with device location variations) of our `FreAuth`.**

I. Introduction

Emerging magnetic wireless power transfer (WPT) technology has significantly alleviated the burden of charging portable devices. This technology has found applications in various domains, including consumer electronics, vehicles, medical devices, and Internet of Things (IoT) [1]–[3]. Magnetic WPT makes it more convenient to charge devices, but it also poses security risks. One of the most obvious threats is freeloading [4], where anonymous devices can access a private wireless charger simply by approaching it [5]. Additionally, there are illegal device attacks that are even more serious. Attackers can damage exposed transmitters (TXs) by inducing high voltage and current through abnormal resonance attack [6], [7]. They may also eavesdrop on charging devices [8], or hijack the communication channel by injecting malicious messages to gain control over the entire charging process [5], [9]. Therefore, it is essential to identify and authenticate legitimate devices in magnetic WPT systems to prevent illegal access and ensure smooth usage.

Currently, the most commonly used authentication method involves the receiver (RX) device uploading its identity (ID) through the communication protocol, such as the Qi Standard [10]. However, this authentication information is vulnerable to forgery as electromagnetic signals are exposed, and the communication protocol is unencrypted [5].

Hardware fingerprint-based device authentication is popular for wireless communication modules [11]. There are two categories: 1) extracting device-specific characteristics from electromagnetic signals emitted due to imperfections in the frequency generator [12], but it lacks permanence since the extracted features are vulnerable to environmental conditions; 2) using temporal features such as oscillator difference caused clock skew [13], [14], but they are unstable and require a long authentication process.

When it comes to magnetic WPT systems, we cannot simply adopt the authentication methods used for wireless communication modules, because these modules are typically expensive and may not be available for low-cost IoT devices. After carefully evaluating the WPT implementation circuits, we have observed that each RX has its own frequency-related features, i.e., the impedance of each electrical component on the RX circuit changes at different working frequencies. We believe that this observation presents an opportunity to use frequency features as a hardware fingerprint for device authentication. Despite its attractiveness, this method has not been thoroughly studied yet, and there are following three technical challenges that should be addressed.

- **Support for RX-independent measurements.** Magnetic WPT systems can use either in-band (e.g., Qi [10], [15]–[18]) or out-band (e.g., BLE and NFC) communication mechanisms. However, these mechanisms may not always be available in certain application scenarios, especially for low-cost IoT devices. Therefore, device authentication should be independent of the RX, which means that impedance measurement and fingerprint extraction should be performed at the TX side without any cooperation or communication from the receiver.
- **Elimination of ideal RX impedance.** Even if we can acquire the RX-related impedance at the TX side, it

is still mainly influenced by the ideal RX impedance and payloads. These terms depend on the ideal circuit elements and the operational state of the device, which are unknown at the TX side. As a result, it is non-trivial to remove these ideal terms and retain the rather weak frequency features.

- **Robustness to environment variations.** In a real-world application, the measurement results (i.e., the RX-related impedance) can be easily influenced by changes in TX-RX coupling caused by environmental variations, such as changes in RX location. Obtaining a stable hardware fingerprint is crucial, and eliminating this interference is essential. However, this task is challenging because the variation in TX-RX relative location is unpredictable.

To this end, in this paper, we will delve into the RX device authentication problem in magnetic WPT systems and endeavor to tackle the aforementioned technical challenges. Instead of relying on extra wireless communication modules, we propose `FreAuth`, a reliable (i.e., high-accuracy), stable (i.e., time-invariant), and robust (i.e., anti-interference) Frequency feature-based device Authentication mechanism in magnetic WPT systems. The main contributions of this paper are summarized as follows.

- **Original authentication approach.** Our `FreAuth` is a novel work to investigate the device authentication problem based on frequency features in magnetic WPT systems. The proposed method is completely RX-independent, i.e., without requiring communication or cooperation from the receiver.
- **Novel hardware fingerprint extraction.** We propose `FreAuth`, a reliable hardware fingerprinting solution for device authentication. By estimating the RX-related impedance at the TX side, we achieve RX-independent measurements and use an interleaved subtraction strategy to remove the ideal RX impedance. Environmental variations are treated as a constant factor and normalized. Finally, our solution uses a Fréchet distance-based algorithm for fingerprint matching during authentication.
- **Prototype implementation and comprehensive evaluation.** We test our `FreAuth` on over 60 devices and achieve an accuracy rate of 95.74%. The extracted fingerprints remain stable for up to 20 days and resist variations in RX payload and location.

II. FreAuth Design

A. Overview

The proposed system comprises two phases, i.e., enrollment, and authentication. In the enrollment phase, the fingerprints of approved devices are collected and stored in a fingerprint library. In the authentication phase, the fingerprints of the devices are extracted and compared with the stored fingerprints in the library for authentication.

The architecture of the proposed system `FreAuth` is shown in Fig. 1. It can mainly be divided into five steps: *Relative Impedance Estimation*, *Ideal Impedance Elimination*, *Mutual Inductance Elimination*, *Fingerprint Database Establishment*, and *Device Authentication*. In the following subsections, each part will be explained in detail.

B. Relative Impedance Estimation

As low-cost IoT devices may lack communication capability, an RX-independent approach is necessary to collect raw data and extract frequency features at the TX side. Therefore, we define and calculate relative impedance in this section.

1) Relative Impedance Definition: Based on Kirchhoff's circuit law, the circuit equations that describe the system shown in Fig. 2 are expressed as:

$$\begin{cases} v_T = z_T(\omega) i_T - j\omega m i_R, \\ j\omega m i_T = z_R(\omega) i_R, \end{cases} \tag{1}$$

where ω is the system working frequency. $i_T(i_R)$ denotes the TX(RX) current, v_T is the TX source voltage. m is the mutual inductance between the TX and RX. $z_T(\omega)$ and $z_R(\omega)$ are the impedance of TX and RX under the working frequency ω, i.e., $z_T(\omega) = r_T + j\omega l_T + \frac{1}{j\omega c_T}$ and $z_R(\omega) = r_p + j\omega l_R + \frac{1}{j\omega c_R}$. $r_T(r_p)$, $l_T(l_R)$, and $c_T(c_R)$ denote the resistance, inductance, and capacitance of TX(RX), respectively.

By taking the frequency factor into account, RX impedance can be obtained as follows according to Eq. (1).

$$z_R(\omega) = \frac{\omega^2 m^2 i_T}{v_T - z_T(\omega) i_T}. \tag{2}$$

It is impossible to compute $z_R(\omega)$ directly because m is not available without measurement from a special instrument. Nonetheless, we know that mutual inductance is decided by TX-RX relative location, but independent of working frequency. Thus, we can move m to the left side of the formula, and RX relative impedance $z_R(\omega, m)$ is defined as

$$z_R(\omega, m) \triangleq \frac{z_R(\omega)}{m^2} = \frac{\omega^2 i_T}{v_T - z_T(\omega) i_T}, \tag{3}$$

where v_T (TX voltage), i_T (TX current), and $z_T(\omega)$ (TX impedance) can be obtained without the cooperation of RX.

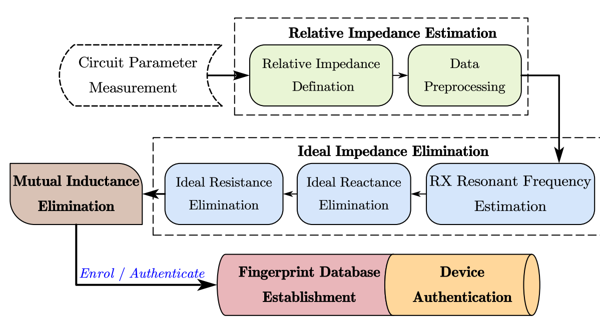


Fig. 1. System structure.

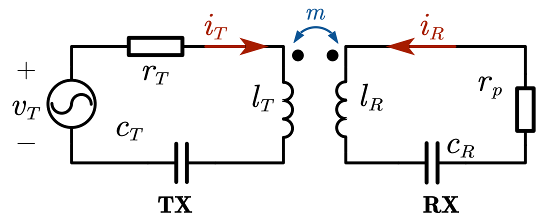


Fig. 2. Schematic of the magnetic WPT system in SISO scenario.

2) Data Preprocessing: We assume $z_T(\omega)$ is already known for each ω, since it can be measured offline with special equipment. To determine the relative impedance across various frequencies, we fix v_T and then measure i_T while the RX is undergoing charging. We set our sampling rate at 1KHz, ensuring that the entire process is completed within one second. This rapid completion allows us to assume that both the location and the payload of the RX remain unchanged during the measurement period. To enhance the data quality by mitigating noise, we incorporate a band-pass filter in our data collection procedure.

As we aim to observe the consecutive characteristic curve across varying frequencies, we utilize the least squares approximation for curve fitting, enabling access to relative impedance at any desired frequency within the range. To demonstrate our `FreAuth`'s effectiveness, we collected three data sets by altering RX location and payload. Figure 3(a) displays the impedance curves. It is evident that while the relative impedance of the RX under different conditions follows a similar trend, it remains distinctly unique. Hence, relative impedance cannot be directly used as a fingerprint.

C. Ideal Impedance Elimination

To extract stable frequency characteristics, we need to eliminate irrelevant variables in RX relative impedance. The relative impedance $z_R(\omega, m)$ can also be expressed as follows.

$$z_R(\omega, m) = \frac{1}{m^2}\Big\{r_p + j\omega l_R + \frac{1}{j\omega c_R} + \sigma(\omega)\Big\}. \quad (4)$$

As for ideal impedance, r_p represents the real part (i.e., the payload), and $j\omega l_R + 1/(j\omega c_R)$ represents the imaginary part. $\sigma(\omega)$ represents the target frequency features, which is quite small as compared with ideal terms. To capture fine-grained frequency features as device fingerprints to uniquely characterize hardware-related disparities between RXs, it is imperative to eliminate the ideal terms. This process will be carried out through the following four steps.

1) RX Resonant Frequency Estimation: The first step is to acquire RX resonance frequency, which can be estimated by finding the frequency where RX impedance is minimum. Because the m is irrelevant to ω, the resonant frequency, denoted as ω_s, can be estimated as follows:

$$\omega_s = \arg\min_{\{\omega\}} |z_R(\omega)| = \arg\min_{\{\omega\}} |z_R(\omega, m)|. \quad (5)$$

Note that, RX impedance will be small under the resonant frequency ω_s, which may results in high current to damage the RX. To avoid this, we configure a low amplitude (i.e., 1V) for v_T during the frequency sweeping process, which is small enough to prevent large RX current. Therefore, our resonant frequency search process will not cause any damage to RX.

2) Dual-Frequency Phenomena Illustration: In the second step, we will introduce the dual-frequency phenomena. As we know, the capacitor and inductor on the RX circuit satisfy $j\omega_s l_R + 1/(j\omega_s c_R) = 0$ under resonant frequency ω_s. By substituting this condition into Eq. (4), we have the following equations for two given frequencies $\alpha\omega_s$ and ω_s/α,

$$\begin{cases} z_R(\alpha\omega_s, m) = \dfrac{1}{m^2}\Big(r_p + j(\alpha - \dfrac{1}{\alpha})\omega_s l_R + \sigma(\alpha\omega_s)\Big), & (6) \\ z_R(\dfrac{\omega_s}{\alpha}, m) = \dfrac{1}{m^2}\Big(r_p - j(\alpha - \dfrac{1}{\alpha})\omega_s l_R + \sigma(\dfrac{\omega_s}{\alpha})\Big), & (7) \end{cases}$$

where $\alpha \in [1, \theta]$ is a constant real number, and the upper bound θ depends on hardware implementation.

Equations (6) and (7) suggest that RX exhibits a conjugate behavior for two dual frequencies, namely $\alpha\omega_s$ and ω_s/α, around the resonance frequency ω_s. This behavior is referred to as the dual-frequency phenomenon.

3) Ideal Reactance Elimination: Since ω_s has already been estimated in the first step, eliminating ideal reactance becomes quite straightforward. We simply add Eqs. (6) and (7), then the result is shown as follows.

$$\begin{aligned} F_{dual}(\alpha, m) &= z_R(\alpha\omega_s, m) + z_R(\frac{\omega_s}{\alpha}, m) \\ &= \frac{1}{m^2}\Big(\underbrace{2r_p}_{\text{irrelevant term}} + \underbrace{\sigma(\alpha\omega_s) + \sigma(\frac{\omega_s}{\alpha})}_{\text{frequency characteristic}}\Big). \end{aligned} \quad (8)$$

$F_{dual}(\alpha, m)$ is still a unique characteristic for device authentication since it contains $\sigma(\alpha\omega_s)$ and $\sigma(\frac{\omega_s}{\alpha})$. Fig. 3(b) depicts the relative impedance curves after eliminating the ideal reactance. The three curves exhibit a notably similar trend in their variations. However, distinctions persist due to the influence of payload (r_p) and mutual inductance (m). Therefore, we need to remove this irrelevant term.

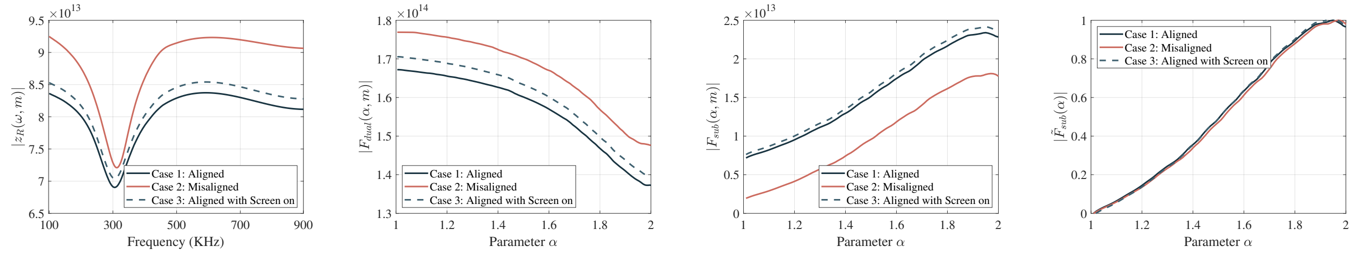
4) Ideal Resistance Elimination: After closely examining Eq. (8), it becomes apparent that the elimination of irrelevant terms can be accomplished by subtracting $F_{dual}(\beta\alpha, m)$ from $F_{dual}(\alpha, m)$, where β is a predetermined coefficient based on the frequency scanning range.

$$\begin{aligned} F_{sub}(\alpha, m) &= F_{dual}(\alpha, m) - F_{dual}(\beta\alpha, m) \\ &= \frac{1}{m^2}\Big(\sigma(\alpha\omega_s) + \sigma(\frac{\omega_s}{\alpha}) - \sigma(\beta\alpha\omega_s) - \sigma(\frac{\omega_s}{\beta\alpha})\Big). \end{aligned} \quad (9)$$

The definition of $F_{sub}(\alpha, m)$ suggests that we can basically eliminate $2r_p$. Thus, it is almost a proper frequency feature related fingerprint except for the unknown TX-RX mutual inductance m. Fig. 3(c) plots the ideal resistance elimination results (i.e., $F_{sub}(\alpha, m)$).

D. Mutual Inductance Elimination

Currently, we have obtained frequency features, in which m is still a factor that is related to the environment. To obtain a fingerprint that is independent of the environment, we must further eliminate it.



(a) Relative impedance (b) Ideal reactance removal (c) Ideal resistance removal (d) m removal

Fig. 3. Process of fingerprint extraction (The TX-RX vertical distance is fixed as 10 mm). Case 1 refers to the TX and RX are aligned; Case 2 denotes the TX and RX are misaligned with 8 mm horizontal distance; Case 3 represents the TX and RX are aligned with keeping the smartphone's screen open.

Given the limited number of sample points, our data collection process will be completed in a relatively short period of time, offering a significant speed advantage over the movement of the RX. Furthermore, the conclusion in [15] suggests that mutual inductance is insensitive to surrounding objects moving. Considering that mutual inductance m is a constant and real number during the measurement process, we can effectively minimize its impact on the fingerprint by normalizing $F_{sub}(\alpha, m)$ as follows

$$\begin{aligned}\widetilde{F}_{sub}(\alpha) &= \text{Norm}(F_{sub}(\alpha, m)) \\ &= \text{Norm}(\sigma(\alpha\omega_s) + \sigma(\frac{\omega_s}{\alpha}) - \sigma(\beta\alpha\omega_s) - \sigma(\frac{\omega_s}{\beta\alpha})).\end{aligned} \tag{10}$$

After eliminating almost all the extraneous variables, we can successfully obtain the stable fingerprint, which is denoted as $\widetilde{F}_{sub}(\alpha)$ and illustrated in Fig. 3(d).

E. Fingerprint Database Establishment

In order to perform fingerprint-based authentication, we are required to establish a database that stores the fingerprints of authenticated devices.

Determination of α. Firstly, we set an upper bound on the variable α, denoted as θ, which depends on the frequency scanning range of the hardware implementation. Eq. (10) indicates that we need to reach the frequencies as $\frac{\omega_s}{\beta\alpha}$ and $\beta\alpha\omega_s$ for one given α. Thus, with a given constant β, we have $\theta \leq min(\frac{\omega_s}{\beta\omega_{min}}, \frac{\omega_{max}}{\beta\omega_s})$, where $[\omega_{min},\ \omega_{max}]$ is the frequency scanning range.

Taking Fig. 3(a) for example, the RX resonance frequency is around 300KHz, and working frequency ranges from 100KHz to 900KHz in our prototype. It can be concluded that $\theta \leq 2$ when choosing $\beta = 1.5$. Thus, we choose $\alpha \in [1,\ 2]$, which is also the input range of $\widetilde{F}_{sub}(\alpha)$.

Storage of Fingerprint. We use the result of $\widetilde{F}_{sub}(\alpha)$ for $\alpha \in [1, \theta]$ as device fingerprint. Because $\widetilde{F}_{sub}(\alpha)$ is a complex number, we store its amplitude and phase separately, i.e., two different curves over variable α.

F. Device Authentication

The proposed fingerprint contains two kinds of curves, i.e., the amplitude and phase ones, and we authenticate devices by matching them. However, measurement noises, mutual inductance fluctuation, and curve fitting error make it impossible for the curves of the same device to be exactly the same. Thus, we adopt a similarity matching method for device authentication (i.e., computing the similarity of amplitude and phase curves, respectively), rather than a direct curve comparison.

We adopt the Discrete Fréchet (DF) Distance-based algorithm as our matching method, and the matching objects are those phase and amplitude curves stored in the database. The general goal of DF is to measure the similarity between two given curves by considering the order in position and parameter α. Hence, DF can ensure that similar curves will have the same shape and features, which fits our requirements.

Phase curves are more stable than amplitude curves. When authenticating a device, we match the phase curves to find candidates and use amplitude curve matching to select the winner. We authenticate the winner only if the similarity level exceeds a preset threshold. A threshold value of 0.2 yielded the best results in our prototype for the test devices, as shown in Table I.

III. Prototype Implementation

As shown in Fig. 4, we have prototyped the proposed `FreAuth` system.

A. TX Side

Our TX side consists of a power transmission circuit, a data collection module and a controller.

Power Transmission Circuit: We employ a uniform copper coil (radius: 25 mm, inductance: 10 μH) alongside three parallel NP0 capacitors (totaling 247 nF) to ensure high stability and achieve a theoretical resonance frequency of approximately 101 kHz. A half-bridge inverter, the LMG5200, and a waveform generator, the AD9833, together constitute a controllable AC power source.

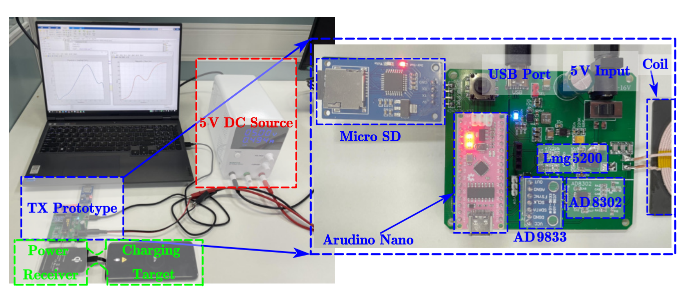


Fig. 4. Prototype of `FreAuth`.

TABLE I
DEVICE LIST OF POWER RECEIVERS AND CHARGING TARGETS

Power Receiver		**Charging Targets**	
Commercial	Self-made	Cellphone	Electronics
Nillkin #2	A # 1	iPhone 5 #2	PHILIPS Mouse #3
FAST #2	B #1	iPhone 6 #2	HP Mouse #2
BASEUS #2	C #2	Xiaomi MX2s #2	Microsoft Mouse #1
YKing #5	D #3	Samsung S8 #2	Earphone #2
MyMax #10	E #3	Huawei Nova5 #2	Resistor #2
PZOZ #10	F #5		
Universal #10	G #5		

Data collection module: We adopt AD8302 to measure TX current as that in [19]. An Arduino Nano takes responsibility for scanning over the frequency range.

Controller: A Lenovo Y9000P laptop equipped with an i7-11800H processor acts as the controller for our `FreAuth` mechanism. The controller communicates with the data collection module to initiate and manage the frequency scanning process.

B. RX Side

"Separated" RXs: In the context of WPT systems, each RX consists of two parts, i.e., the power receiving circuit (denoted as power receiver) and the circuit payload (denoted as charging target). It is important to note that the proposed frequency feature is inherently related to the power receiving circuit and not the payload. This is because, during the charging process, the payload solely exhibits a frequency-independent resistance. Therefore, although these two parts are normally integrated together in commercial products, we still adopt RXs in "Separated" form to better control the factors for thorough evaluation of the `FreAuth` mechanism.

Devices composition: For power receiver part, we have as much as 61 different devices, including self-made and commercial ones. For charging target part, we choose different types of commercial devices, such as smartphone, mouse, *etc*. Table I gives a detailed list of these devices used for evaluation.

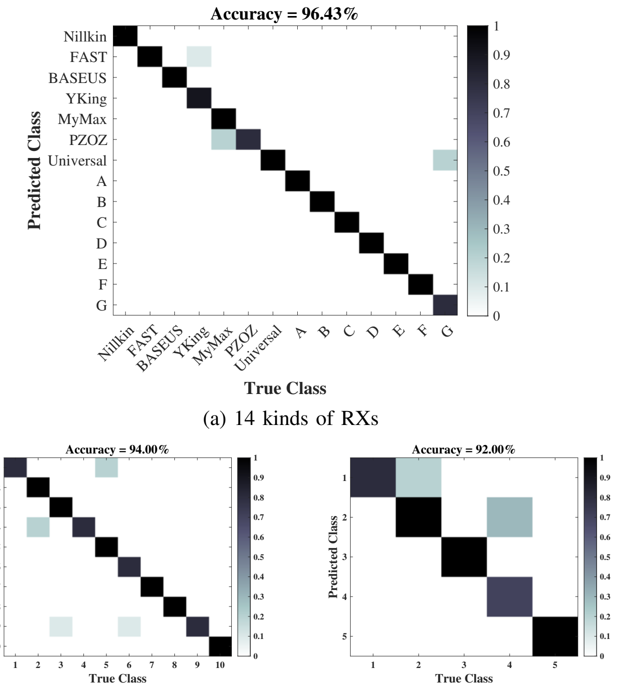
IV. PERFORMANCE EVALUATION

A. Overall Performance

Here, we analyze the authentication accuracy and speed of `FreAuth`, respectively.

1) Authentication accuracy: **Accuracy over different RX models.** We first evaluate the distinguishability of `FreAuth` on different power receiver models. We combine 14 RXs from the devices listed in Table I, i.e., different power receivers are of different models, and each one is combined with randomly chosen charging targets. For each RX, we perform 10 authentication processes among these 14 RXs, i.e., database only containing fingerprints of these 14 devices.

Fig. 5(a) plots the confusion matrix over the 14 RXs. As shown in the figure, `FreAuth` achieves overall authentication accuracy of 96.43%. To be more specific, 5 failed authentication cases, i.e., *YKing* $\rightarrow$ *FAST*(1 times), *MyMax* $\rightarrow$ *PZOZ*(2 times), and *Universal* $\rightarrow$ *Self-made G*(2 times). Although there doesn't exist 100% authentication failure case, we can still improve the accuracy by performing multiple authentications.



(a) 14 kinds of RXs

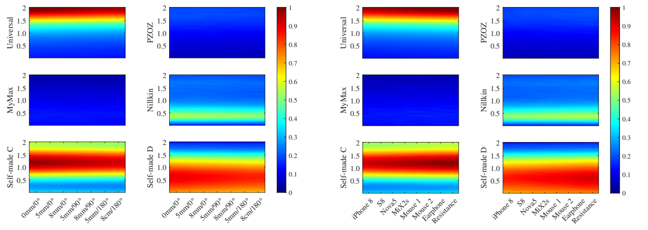
(b) 10 PZOZ RXs (c) 5 Self-made RX G

Fig. 5. The confusion matrix for authentication.

Accuracy over the same model. For a device authentication mechanism to be deemed effective, it must be capable of distinguishing between multiple devices of the same model. Therefore, we evaluate the performance of `FreAuth` in authenticating devices of the same model, i.e.we independently carry out evaluations on 14 different types of power receivers. We select results from two representative power receivers for analysis: *PZOZ*, which represents commercially available power receivers, and *Self-made G*, which represents the dedicated receivers. In our experimental setup, we construct 15 receivers by combining the 15 power receiver instances (10 from *PZOZ* and 5 from *Self-made G*) with different payloads, as shown in Table I. We then conduct 10 authentication processes for each receiver.

As illustrated in Figs. 5(b) and 5(c), we achieve a 94.00% and 92.00% accuracy for *PZOZ* and *Self-made G*, respectively, with no cases of a device failing authentication in all 10 attempts. Since these power receivers share similar hardware structures and `FreAuth` leverages hardware imperfections, we can expect comparable experimental results with other receivers. Therefore, we can conclude that our `FreAuth` demonstrates a remarkable ability to distinguish between instances of the same model.

Overall Accuracy. We finally perform evaluation on the total 61 receivers. To conveniently control payloads of the receivers, we reuse the charging targets in Table I as the payloads for the 41 commercial receivers, and choose the resistors with different values as the payloads for the 20 self-made receivers. For each receiver, we perform authentication



(a) Location (b) Payload

Fig. 6. Stability of fingerprints at different environments.

10 times and take the hit count (i.e., number of successful authentication) as the evaluation metric, where the database contains fingerprints of these 61 RXs.

We achieved an average accuracy of 95.74% among all receivers, which obviously outperforms the reported accuracy of 89.7% for state-of-the-art solution Qid [13], which is designed for identifying Qi-compatible devices through the time interval of data packets.

Note that RXs with a separated pattern (i.e., receiver + payload) are equivalent to their integrated counterparts at the circuit level. Thus, `FreAuth` can ensure authentication accuracy for integrated commercial WPT RXs.

B. Practicability Study

In this section, we evaluate the reliability and robustness of `FreAuth` under real-world deployments. We select 6 representative power receivers (including four commercial ones and two self-made ones) as study objects.

1) Impact of Location: To ensure a device is authenticated regardless of its location, we place six devices at ten random locations. Results in Fig. 6(a) show that the fingerprints are significantly different between devices but show negligible changes with locations. This proves the effectiveness of `FreAuth` in removing the influence of environmental factors.

2) Impact of Payload: We attach six power receivers with randomly selected charging targets in Table I. The fingerprints in Fig. 6(b) remain stable, indicating that `FreAuth` could eliminates the influence of the RX payload.

V. Conclusion

We have introduced `FreAuth`, a new technology that uses frequency features to authenticate devices in magnetic WPT systems. Our experiments indicate that `FreAuth` can authenticate over 60 devices with an accuracy of 95.74%. This makes it an excellent solution for medium-scale wireless charging scenarios. In future work, we aim to streamline fingerprint construction and enhance universality while maintaining accuracy.

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